

SOMADA: A Web-Based Laboratory Management System

Project Documentation Submitted to the Faculty of the
School of Engineering and Technology

NU Fairview

In Partial Fulfillment of the Requirements for
Systems Analysis and Detailed Design for IT
MSYADD1

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I. Project Overview

SOMADA is a web-based laboratory management system for the Physics and Circuits Laboratory of NU Fairview. It replaces the laboratory's paper forms and spreadsheet records with one online system that holds the inventory, room bookings, borrowing records, clearances and disposal records together and keeps them consistent.

The laboratory is a teaching facility on the fourth floor of the campus, open from 8:30 AM to 5:30 PM, serving the laboratory sessions of physics, circuits and information technology subjects. It holds 322 catalogued items in eight categories. Some are durable equipment such as oscilloscopes, multimeters, Arduino kits and physics apparatus. The rest are consumable supplies such as resistors, jumper wires, soldering lead and batteries. One person, the laboratory head, manages all of it.

Every transaction currently begins on paper. Students and faculty fill out borrower's slips and laboratory service request forms by hand. The laboratory head reviews each one, then types the same information again into a protected Google Sheets workbook of eleven worksheets. Room availability is checked against a separate weekly grid for each room, and on those grids the difference between a class session, a lecture booking and a special request exists only as the laboratory head's own colour coding. Approvals are signatures collected in person, and the signed forms are kept outside the workbook.

None of these records are connected to each other, and that is where the difficulty starts. The borrowing records are not linked to the inventory, so one mistyped quantity quietly changes every figure that depends on it, and nobody can tell how much of a consumable is left without counting it by hand. Approvals are slow because a form has to be carried to a person for signature. Scheduling conflicts turn up only after they have happened. A shortage of supplies is discovered at the moment a scheduled session needs the item. Students and faculty who want to know whether an item is available, or whether their request has been approved, have to visit the laboratory or wait for the laboratory head to reply. When an

instrument comes back damaged, or does not come back at all, the only way to find out who is accountable is to dig up the signed paper slip from that session.

The proposed system puts all of this on one platform. A student or faculty member files a laboratory service request at a kiosk in the laboratory, picking the usage type, choosing items from a catalogue that shows live stock and picking the room, date and time from the schedule on screen; a one-time request holds a single time block, while a whole-term request holds the corresponding block in each remaining week of the term, and the system checks every block it would hold for conflicts before it accepts the request. The assigned professor or the Dean then approves by selecting a button inside an automated email, so no signature and no second account are needed. From the approved request, the laboratory head issues the equipment, and the system builds the borrowing record, deducts the stock and reconciles the returns after the session, raising a clearance against the accountable student where an item comes back broken or does not come back at all. Items withdrawn from the laboratory for good go into a disposal log. Three functions use artificial intelligence: a forecast that flags consumables likely to run out before an upcoming session, a virtual assistant that answers students and faculty on equipment availability, schedules, operating hours. The assistant accepts a question only from a requester it can identify in the laboratory's own records, a student by NU Student ID or a faculty member by NU Fairview email address. The laboratory head holds the only account, and students follow their requests on a tracking screen that needs only the request reference. At the end of the term the head exports the report of laboratory activity and submits it to the administration. Payment for lost items, outside repairs and the hand-off to the registrar stay manual.

II. Methodology

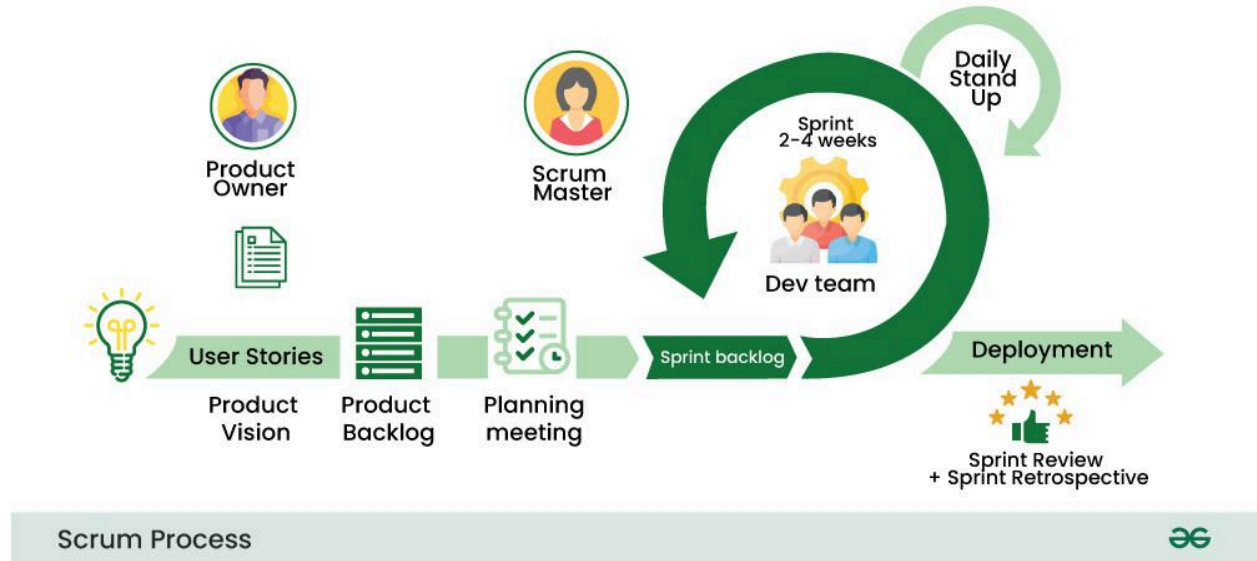


Figure 1: Scrum Framework

The system is developed under the Agile methodology, following the Scrum Framework. Agile was selected because the requirements of this project continued to change throughout requirements gathering.

This project spans three (3) terms, each term running thirteen and a half (13.5) weeks and corresponding to one subject and one release: MNTSDEV in the second year third term, MYSADD1 in the third year first term, and MCSPROJ in the third year second term.

Product Vision. The product vision was formed during MNTSDEV from the operational problems of the Physics and Circuits Laboratory and the Product Owner: paper borrower's slips and laboratory service request forms re-encoded by hand, a Google Sheet and several workbooks whose borrowing records are not linked to its inventory, room availability checked against a separate weekly grid for each room, approvals signed on paper and stored outside the workbook, and consumable quantities that cannot be established without a physical count.

Product Backlog and User Stories. From that vision the team wrote the objectives as user stories and organized them into the product backlog presented in Table 1: to digitalize inventory and equipment borrowing, to provide reservation with conflict detection, to replace paper-based clearance, to record approval through an emailed confirmation control and publish request status, to add demand forecasting and a virtual assistant, and to record waste disposal. The backlog is a living artifact and is refined continuously as consultation produces new information.

Sprint Planning. At the start of each sprint, the team will pull the backlog items it forecasts it can complete into sprint backlogs, and state a sprint goal. However, it may be adjusted if there is new information or miscalculations.

The Sprint. Sprints will run at a fixed length. However, the fixed length will be decided when the development phase starts.

Sprint Review and Sprint Retrospective. Each sprint closes with a sprint review, at which is demonstrated to the laboratory head and the backlog is adjusted in light of what he raises, and then a sprint retrospective, at which the team inspects how it worked during the sprint, its collaboration, its tools and its process, and commits to at least one improvement for the next sprint. The review inspects the product, while the retrospective inspects the process.

Product Backlog Refinement. Findings from the review and the retrospective return to the product backlog, which the Product Owner reorders before the next sprint planning.

Current Status. As of this submission the product vision, the product backlog, the requirements analysis, the system models and the prototype are complete or in revision, and construction has not begun. Every item in Table 1 therefore carries the status Not Started, and the fixed length of the sprint is left unassigned until the development phase starts.

III. Requirements Analysis

3.1.1 Product Backlog

Table 1: Product Backlog

ID	As a	I want to be able to	So that	Priority	Sprint	Status
01	Laboratory Head	add, edit, or remove equipment records	I can accurately manage and categorize what is currently in the storage room.	Must	1	Not Started
02	Laboratory Head	log borrowed items that record student ID numbers and time	I can prevent mistakes caused by manual typing or handwriting.	Must	3	Not Started
03	Laboratory Head	record a broken or missing item against the accountable group	I can document damaged items and clearly mark the responsible person.	Must	3	Not Started
04	Laboratory Head	View a live screen showing item availability and return times	I can track exactly how many items are left and when they are expected to be returned.	Should	3	Not Started
05	Laboratory Head	generate printable summaries and transaction logs	I can review equipment usage patterns, identify busy hours, and submit required end-of-term reports.	Should	4	Not Started
06	Laboratory Head	generate a digital clearance form	I can record accountability for an unreturned or broken item and clear the student before the end of the term.	Must	3	Not Started

07	Student / Faculty	Select the equipment required on the laboratory service request form	I can state what the session needs online without filling out a separate paper slip.	Must	2	Not Started
08	Student / Faculty	View a visual calendar and request a room	I can secure a laboratory space without conflicting with others' schedules.	Must	2	Not Started
09	Faculty / Dean	confirm an equipment or room request from an emailed link without signing in	I can grant permission without a signature or an account, and the request is not held up when I am unavailable in person	Must	2	Not Started
10	Student / Faculty	request a vacant time block as an out-of-schedule laboratory reservation at least two days in advance	I can complete a missed laboratory session or book the facility for a defense without going through the laboratory head in person	Must	2	Not Started
11	Laboratory Head	have whole-term requests remain open across the term while one-time requests close on completion	I can see at a glance how many requests are actually finished without recurring ones distorting the count	Must	2	Not Started
12	Laboratory Head	record issued and returned quantities separately, with equipment barred from being marked consumed	consumed counts reflect only what was genuinely used up, and unreturned equipment is correctly recorded as broken or lost	Must	3	Not Started

13	Laboratory Head	log items withdrawn for disposal and flag those that are hazardous waste	I can show what left the laboratory and why, and keep hazardous items separated for proper collection	Should	3	Not Started
14	Laboratory Head	see forecasted consumable demand and be alerted to items likely to run out before a scheduled session	I can restock ahead of time instead of discovering the shortage during the session	Must	4	Not Started
15	Student / Faculty	ask a virtual assistant about item availability, schedule availability, and operating hours	I can get answers outside office hours without visiting the laboratory in person	Must	4	Not Started
16	Laboratory Head	receive automatically generated insights from the dashboard metrics	I can act on usage patterns without interpreting the raw figures myself	Should	4	Not Started

3.1.2 Event Tables

Table 2: Event Tables

EVENT	TRIGGER	SOURCE	USE CASE	RESPONSE	DESTINATION
Laboratory head signs in	Head enters NU Fairview credentials on the login screen	Laboratory Head	Log in to System	Credentials verified and administrative screens unlocked	Main Dashboard
Requester files a laboratory service request	Requester completes the request form at the laboratory kiosk and submits	Student, Faculty	File Laboratory Service Request	Request saved as Pending and made available for approval	SR Records, Requester
Approver confirms a request	Professor or Dean selects the confirmation control in the emailed notification	Faculty, Dean	Approve Service Request	Approval request email dispatched to the approver; on the approver's response the status changes from Pending to Approved or to Declined	SR Records, Approval Records, Professor/Dean email, Requester tracking
Head issues equipment for a session	Head opens an approved request and records the quantity issued per item	Laboratory Head	Issue Equipment	Borrowing record created against the requesting faculty, the group members and the date; issued quantities deducted from stock	Inventory, SR Records
Session ends and items are returned	Head records returned quantities against quantities issued	Laboratory Head	Process Return	Quantities reconciled; unreturned consumables marked consumed;	Inventory, Broken/Loss/Consumed Records

				equipment not returned in working order marked broken or lost	
Accountability is raised for a damaged or missing item	Head selects the accountable group from the borrowing record	Laboratory Head	Process Clearance	Clearance record created against the accountable student and tracked to settlement	Clearance Records, Student tracking
Item is withdrawn from the laboratory	Head records the item, reason, quantity and classification	Laboratory Head	Log Waste Disposal	Disposal entry created; hazardous items flagged for separate collection	Waste Disposal Log, End-Term Report
Head maintains the equipment catalogue	Head adds, edits or removes a holding in Inventory Management	Laboratory Head	Manage Equipment Inventory	Catalogue updated and availability recomputed; a low stock flag is raised where a quantity falls below its reorder level	Equipment Inventory Records, Dashboard
Head reviews forecasted consumable demand	Head opens the dashboard and the system recalculates over the historical usage and consumption records	Laboratory Head	Forecast Consumable Demand	Items at risk listed with projected depletion date	Dashboard, Laboratory Head
Requester checks the status of a request	Requester enters a reference number on the tracking screen	Student, Faculty	Track Request Status	Status displayed as Pending, Approved, Declined, Ongoing or Completed	Requester
User queries	User submits	Student,	Query Virtual	Assistant	Requester

the assistant	a question	Faculty	Assistant	answers on item availability, schedule availability, and operating hours	
Head opens the dashboard	Head signs in and dashboard metrics load	Laboratory Head	View Real-Time Inventory & Dashboard	Metrics summarized and automated insights generated	Dashboard, Laboratory Head
Term ends	Head triggers the end-of-term report	Laboratory Head	Generate End-Term Report	Usage per faculty, most-borrowed items and broken, lost and consumed totals compiled and exported to the laboratory head for submission	Laboratory Head, then NU Fairview administration
Dean's office issues the term schedule	Head records the committed and vacant blocks for each laboratory room	Laboratory Head	Manage Laboratory Logs & Schedule	Laboratory schedule updated for each room and vacant blocks published for out-of-schedule requests	Laboratory Schedule
Working day ends	Head completes and submits the daily task report	Laboratory Head	Manage Daily Task Log	Daily task entry stored against the date and made available to reporting	Daily Task Logs, End-Term Report
Head records the usage of a completed session	The head has reconciled the return for the session	Laboratory Head	Manage Laboratory Logs & Schedule	Usage entry recorded in the Physics or Circuits laboratory log for the session and the borrowing raised against it	Physics and Circuits Laboratory Logs, End-Term Report

3.1.3 Use Case Diagrams

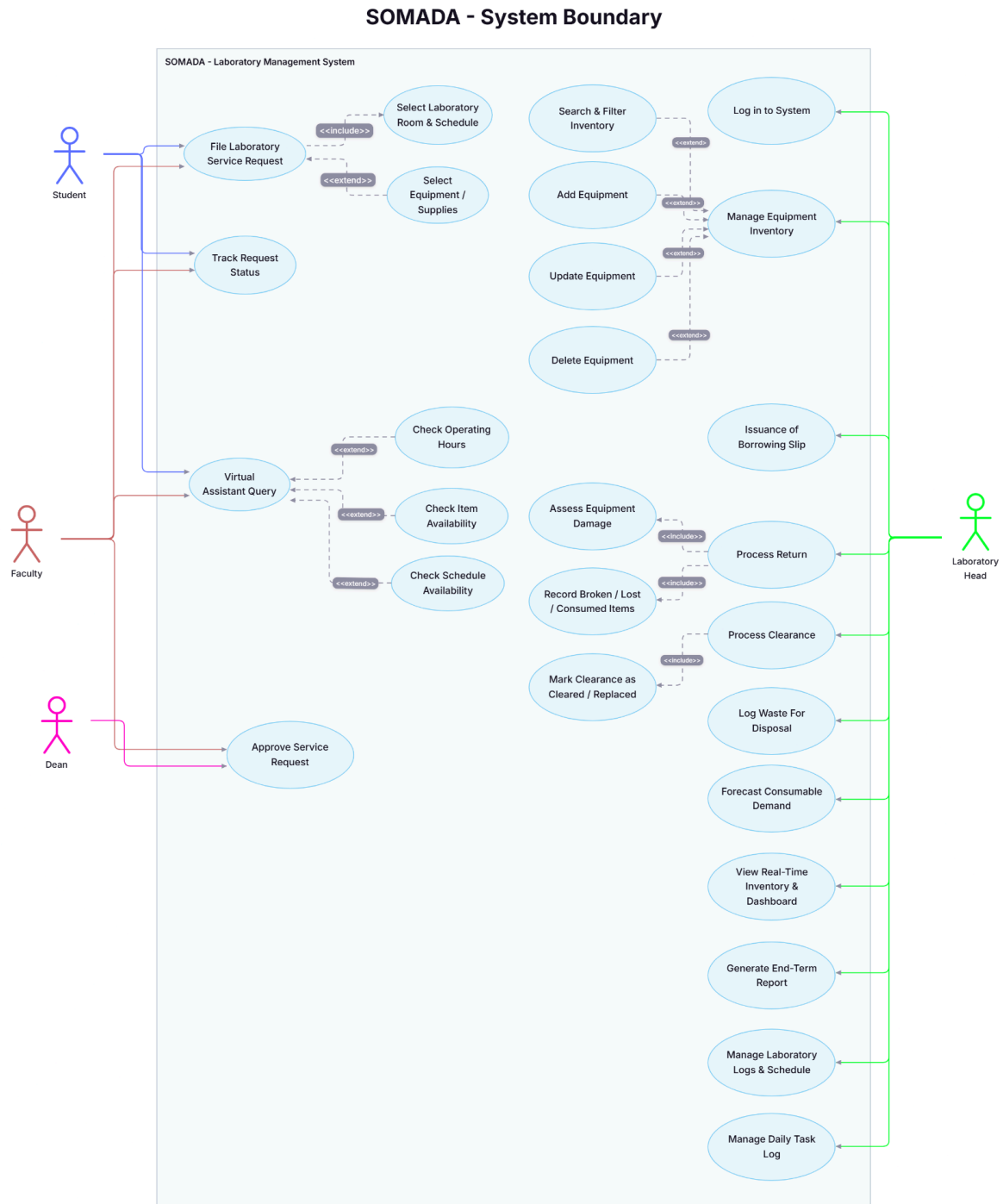


Figure 1: Use Case Diagram

Figure 1 presents the use case diagram. It shows four actors and the fifteen base use cases in which they participate, together with fourteen further use cases reached from those base cases through include and extend relationships. Every event listed in Table 2 is raised by one of the four actors; the system performs no function on a schedule of its own, and the forecast is recalculated when the laboratory head opens the dashboard.

3.1.4 Use Case Full Description

Tables 3 to 17 give the full description of the fifteen base use cases shown in Figure 1.

The included and extended use cases are described within the flow of activities of the base use case to which each belongs.

Table 3: Log in to System

Use Case Name:	Log in to System
Scenario:	The laboratory head needs access to the administrative screens of SOMADA.
Triggering Event:	The laboratory head selects “Log In” and enters NU Fairview credentials.
Brief Description:	The head submits credentials on the login screen. The system verifies them and, on success, establishes a session, applies role-based access and loads the Main Dashboard. The laboratory head holds the only system account; students, faculty and the Dean interact with the system without accounts.
Actors:	Laboratory Head
Related Use Cases:	View Real-Time Inventory & Dashboard, Manage Equipment Inventory
Stakeholders:	Laboratory Head: requires sole controlled access to the administrative screens. NU Fairview: requires laboratory records restricted to authorised personnel. Students and Faculty: rely on records maintained under a single accountable account.

Preconditions:	The laboratory head account exists in the system. The system is reachable from the university network.
Postconditions:	A session is established and the administrative screens are unlocked. The Main Dashboard is displayed. If verification fails, no session is created and no screen is unlocked.
Flow of Activities:	
Actor	System
1. Head opens the login screen. 2. Head enters the NU Fairview credentials. 3. Head submits the credentials.	1. System validates the credentials against the stored account. 2. System establishes the session and applies role-based access. 3. System loads the Main Dashboard.
Exception Conditions:	If the credentials are invalid, the system reports the failure and no session is created. If the account is unavailable, the system displays a failure message. If the head has forgotten the password, the head selects Forgot password on the login screen and a new password is sent to the recorded NU Fairview email address. If the request originates outside the university network, the system refuses access to administrative screens.

Table 4: File Laboratory Service Request

Use Case Name:	File Laboratory Service Request
Scenario:	A student or faculty member needs a laboratory room, equipment, or both, for a session.
Triggering Event:	The requester completes the laboratory service request form at the laboratory kiosk and submits it.
Brief Description:	The requester selects any items required for the session from the catalogue, which carries live

	stock indicators, then selects the laboratory room and the date and time of use from the schedule displayed. The requester enters the Student ID of each group member taking part in the session. The requester states the request type and the basis of the request, on-schedule or out-of-schedule, and states whether the session is for facility use only or for laboratory use requiring equipment. A one-time request holds the single block selected; a whole-term request holds the corresponding block in each remaining week of the term. The system enforces the two-day advance requirement for an out-of-schedule request, checks every block the request would hold against the laboratory schedule, records the request as Pending, issues the request reference, and makes it available for approval by the assigned professor or the Dean.
Actors:	Student, Faculty
Related Use Cases:	Approve Service Request, Track Request Status, Issue Equipment, Manage Laboratory Logs & Schedule
Stakeholders:	Student and Faculty: need a room and equipment secured without completing paper forms. Laboratory Head: needs requests to arrive complete and free of conflicts. Dean: needs out-of-schedule use of the facility to remain visible.
Preconditions:	The laboratory kiosk is available. The Dean's term schedule has been recorded in the laboratory schedule. The requester can identify the assigned professor or the Dean as approver.
Postconditions:	The request is stored with the status Pending. The request becomes available for approval. Every block the request holds is marked Held against further requests. If validation fails, nothing is stored. The declared group members are stored against the request.
Flow of Activities:	

Actor	System
1. Requester opens the laboratory service request form at the kiosk and browses the equipment catalogue. 2. Requester selects the items required for the session and the quantity of each. 3. Requester enters the Student ID of each group member. 4. Requester selects the laboratory room and the date and time of use from the schedule displayed. 5. Requester states the request type, whole-term or one-time, and the basis of the request, on-schedule or out-of-schedule. 6. Requester states the usage type, facility use only or laboratory use requiring equipment. 7. Requester completes the remaining request details and confirms the selected slot. 8. Requester reviews the request and submits the form.	1. System displays the equipment catalogue with live stock indicators drawn from the inventory records. 2. System validates the requested quantities against available stock. 3. System records the declared group against the request. 4. System displays the laboratory schedule for the selected room together with the blocks it leaves vacant. 5. System records the request type and the basis of the request. 6. System records the usage type. 7. System validates the advance notice, applying the two-day requirement to an out-of-schedule request. 8. System checks the selected slot against the laboratory schedule and confirms it. 9. System stores the request as Pending, issues the request reference, and makes it available for approval.
Exception Conditions:	If the block is already committed, the system reports the conflict and offers the remaining vacant blocks. If an out-of-schedule request is filed less than two days in advance, the system rejects it. If a requested quantity exceeds the available stock, the system flags the item so the requester can adjust it. If required information is incomplete, the system requests corrections before submission. If a Student ID is malformed or omitted, the system requests correction before submission.

Table 5: Approve Service Request

Use Case Name:	Approve Service Request
Scenario:	A filed laboratory service request awaits the decision of the assigned professor or the Dean.

Triggering Event:	The system dispatches the approval request email and the approver selects the confirmation control it carries.
Brief Description:	The approver receives an automated email containing the request details and a confirmation control. Selecting the control records approval or decline without a signature and without a system account. The request moves from Pending to Approved or Declined and the requester is notified.
Actors:	Faculty, Dean
Related Use Cases:	File Laboratory Service Request, Issue Equipment, Track Request Status
Stakeholders:	Faculty: must authorise sessions supervised by them without attending in person. Dean: must authorise use of the facility for non-laboratory purposes. Requester: needs the request settled without physically routing a form. Laboratory Head: needs a recorded approval before issuing equipment.
Preconditions:	A request exists with the status Pending. The approver's email address is recorded against the request. The confirmation control has not expired or been used.
Postconditions:	The request status is updated to Approved or Declined. An approval record is written carrying the decision, the approver and the time of decision. The outcome is published to the requester and becomes visible on the request tracking screen. An approved request becomes available to the laboratory head for issuance.
Flow of Activities:	
Actor	System
1. Approver opens the notification email. 2. Approver reviews the request details.	1. System sends the notification containing the request details and the confirmation control.

3. Approver selects Confirm or Decline.	2. System validates the confirmation token. 3. System updates the request status and writes the approval record. 4. System publishes the outcome to the request tracking screen.
Exception Conditions:	If the confirmation token has expired or has already been used, the system refuses it and issues a fresh notification. If the request is declined, the status is set to Declined and the held block is released. If the approver does not respond before the session date, the request remains Pending and the laboratory head follows it up outside the system.

Table 6: Issuance of Borrowing Slip

Use Case Name:	Issue Equipment
Scenario:	An approved laboratory service request requires equipment to be released for the session.
Triggering Event:	The laboratory head opens an approved request and records the quantity issued for each item.
Brief Description:	The laboratory head generates the borrowing record from the approved service request, with the item list constrained to the approved items. The group members declared on the request are presented for confirmation and may be amended to reflect who is present for the session. The quantity issued is recorded per item against the requesting faculty member, the group members and the date of the session, and the issued quantities are deducted from available stock. A borrowing slip is produced and signed by both parties on release.
Actors:	Laboratory Head
Related Use Cases:	Approve Service Request, Process Return, Manage Equipment Inventory
Stakeholders:	Laboratory Head: needs issuance recorded against the inventory without re-encoding. Faculty: needs the session supplied with the

	approved items. Students: need accountability recorded against the correct group.
Preconditions:	The request holds the status Approved. The head is signed in. The approved items are physically available in the storage room.
Postconditions:	A borrowing record is created against the request, the faculty member, the group members and the date. Available stock is decremented by the issued quantity for each item. The request status is set to Ongoing.
Flow of Activities:	
Actor	System
1. Head opens the approved request. 2. Head confirms the items and records the quantity issued for each. 3. Head releases the prepared items and produces the borrowing slip.	1. System displays the item list constrained to the approved items. 2. System validates the issued quantity against the stock on hand. 3. System writes the borrowing record and decrements available stock. 4. System sets the request status to Ongoing.
Exception Conditions:	If an approved item cannot be supplied in the quantity requested, the head records a reduced quantity or a substitute coordinated with the assigned professor. If an item is unavailable, the system blocks issuance of that line. If the request does not hold the status Approved, the system refuses issuance.

Table 7: Process Return

Use Case Name:	Process Return
Scenario:	A session has ended and the borrowed items are

	checked back in.
Triggering Event:	The laboratory head records the returned quantities against the quantities issued.
Brief Description:	Returned quantities are reconciled against issued quantities. Consumables returned unused are not marked consumed; only the unreturned balance is marked consumed and attributed to the borrowing party. Equipment that cannot be returned in working order is marked broken, and equipment not returned at all is marked lost. Equipment is never eligible to be marked consumed. Resulting entries are written to the Broken, Loss and Consumed log and inventory balances are updated.
Actors:	Laboratory Head
Related Use Cases:	Issue Equipment, Process Clearance, Log Waste Disposal, Manage Equipment Inventory
Stakeholders:	Laboratory Head: needs consumed counts to reflect genuine depletion only. Students: need returns credited so that no accountability is recorded in error. NU Fairview: needs losses attributed and reportable at the end of the term.
Preconditions:	A borrowing record exists with the status Ongoing. The returned items have been physically checked.
Postconditions:	Returned quantities are recorded against the borrowing record. Unreturned balances are classified as consumed, broken or lost. The Broken, Loss and Consumed log is updated. Available stock is restored for equipment returned in working order; equipment returned broken remains in the existing quantity with its condition status changed, and is deducted only if it is later withdrawn for disposal; consumed quantities are deducted from the existing quantity at this point and are not deducted again. The request is set to Completed where no balance remains outstanding, and the session becomes available for a laboratory usage entry.

Flow of Activities:	
Actor	System
1. Head checks the returned items after the session. 2. Head records the returned quantity for each item. 3. Head classifies unreturned equipment as broken or lost.	1. System displays the quantities issued for the borrowing record. 2. System computes the unreturned balance for each item. 3. System marks unreturned consumables as consumed and bars equipment from being marked consumed. 4. System writes the entries to the Broken, Loss and Consumed log and updates inventory balances. 5. System sets the request to Completed where no balance remains outstanding.
Exception Conditions:	If a returned quantity exceeds the quantity issued, the system rejects the entry. If equipment is marked consumed, the system refuses the classification and requires broken or lost. If a balance remains outstanding, the request stays Ongoing and a clearance is raised against the accountable group.

Table 8: Process Clearance

Use Case Name:	Process Clearance
Scenario:	An item is found broken or missing during the post-session check and accountability must be recorded.
Triggering Event:	The laboratory head selects the accountable group from the borrowing record.
Brief Description:	Damage or loss is identified during the post-session check rather than reported separately by the students. The head identifies the accountable group from the borrowing record, raises a clearance record against the accountable student stating the item, quantity and reason, and tracks it to settlement, marking it cleared or replaced. Where the borrowing record named no group members, the clearance is raised against the

	requesting faculty member instead. The system issues the clearance copy to the accountable party. Students do not file clearance themselves.
Actors:	Laboratory Head
Related Use Cases:	Process Return, Track Request Status, Generate End-Term Report
Stakeholders:	Laboratory Head: needs accountability attached to the borrower rather than only to the supervising professor. Students: need to be told what is recorded against them and how to settle it. Registrar: receives the settled clearance record from the laboratory head outside the system. NU Fairview: needs unsettled accountability visible at the end of the term.
Preconditions:	A borrowing record identifies the group members for the session. An item is recorded as broken or lost against that record.
Postconditions:	A clearance record is created against the accountable student, or against the requesting faculty member where no group members were recorded. The clearance copy is issued by the system to the accountable party. The clearance is marked cleared or replaced on settlement.
Flow of Activities:	
Actor	System
1. Head opens the borrowing record for the session. 2. Head selects the accountable group member. 3. Head raises the clearance record stating the item, quantity and reason. 4. Head marks the clearance cleared or replaced on settlement.	1. System displays the group members recorded on the borrowing record. 2. System writes the clearance record against the selected student. 3. System records the clearance against the selected student. 4. System updates the clearance status and closes it on settlement.
Exception Conditions:	If no group members were recorded, the clearance is raised against the requesting faculty member.

	If a replacement item differs from the original, it is recorded as a replacement with remarks. If a clearance remains unsettled at the end of the term, it stays open and is listed in the end-term report.
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Table 9: Log Waste Disposal

Use Case Name:	Log Waste Disposal
Scenario:	An item is withdrawn permanently from the laboratory and must be recorded and classified.
Triggering Event:	The laboratory head records the item, reason, quantity and classification in the disposal log.
Brief Description:	Items already recorded as broken, lost or consumed are reviewed to determine which are to be withdrawn permanently; items that can be repaired are retained and are not entered. Each withdrawn item is logged with its reason, quantity, unit and date, the method by which it is disposed of and the name of the person who handled it, and is classified as general or hazardous waste. A disposal entry is tracked as Pending, Scheduled or Disposed. Hazardous items such as spent batteries, soldering lead waste, damaged light bulbs and discarded electronic components are held separately pending collection. The log is linked to the inventory records.
Actors:	Laboratory Head
Related Use Cases:	Process Return, Manage Equipment Inventory, Generate End-Term Report
Stakeholders:	Laboratory Head: needs a traceable record of what has left the facility and why. NU Fairview: needs hazardous waste consolidated for institutional compliance reporting. Future Researchers: rely on the disposal record for end-of-term figures.
Preconditions:	The item appears in the Broken, Loss and Consumed log or in the inventory.

	The head is signed in.
Postconditions:	A disposal entry is created carrying reason, quantity, unit, date, disposal method, handler and classification, and is tracked to completion. The existing quantity is reduced for items still counted in inventory at the time of withdrawal, that is, equipment previously marked broken and items withdrawn directly from stock. Quantities already deducted as consumed or lost are not deducted again. Hazardous entries are flagged for separate collection. The entry is available to the end-term report.
Flow of Activities:	
Actor	System
1. Head reviews the items recorded as broken, lost or consumed. 2. Head selects an item for withdrawal and states the reason and quantity. 3. Head classifies the item as general or hazardous waste. 4. Head saves the disposal entry.	1. System displays the candidate items from the Broken, Loss and Consumed log. 2. System validates the quantity against the recorded balance. 3. System writes the disposal record and reduces the inventory. 4. System flags hazardous entries for separate collection and reporting.
Exception Conditions:	If the quantity exceeds the recorded balance, the system rejects the entry. If the item can be repaired, it is retained and no disposal entry is created. If the classification is omitted, the entry cannot be saved.

Table 10: Manage Equipment Inventory

Use Case Name:	Manage Equipment Inventory
Scenario:	The laboratory head maintains the catalogue of laboratory holdings.
Triggering Event:	The laboratory head opens Inventory Management to add, edit, remove or search a holding.
Brief Description:	Each holding is recorded by name together with its description, category, unit of measure, quantity, condition status, reorder level and an optional

	photograph, with quantities tracked as existing, available, in use and reserved. The distinction between equipment and consumables is enforced: equipment is returnable and cannot be recorded as consumed, while consumables are depletable. The system raises a low stock flag when a quantity falls below its reorder level.
Actors:	Laboratory Head
Related Use Cases:	Issue Equipment, Process Return, Forecast Consumable Demand, Log Waste Disposal
Stakeholders:	Laboratory Head: needs stock levels visible at a glance and changes attributable rather than silently overwritten. Faculty: rely on accurate availability when planning experiments. Students: rely on the catalogue reflecting what can actually be issued.
Preconditions:	The head is signed in. The item category and unit of measure are defined.
Postconditions:	The holding is created, amended or removed. Existing, available, in use and reserved quantities are recomputed. A low stock flag is raised where a quantity falls below its reorder level. The change is attributable to the head and the date.
Flow of Activities:	
Actor	System
1. Head opens Inventory Management. 2. Head searches or filters the catalogue. 3. Head adds, edits or removes a holding. 4. Head saves the change.	1. System displays the catalogue with existing, available, in use and reserved quantities. 2. System applies the search or filter. 3. System validates the entry against the category, unit and type rules. 4. System stores the change and recomputes availability. 5. System raises a low stock flag where a quantity falls below its reorder level.

Exception Conditions:	If removal is attempted for an item with an outstanding issued quantity, the system refuses it. If a consumable is marked returnable or equipment is marked depletable, the system rejects the entry. If a duplicate name exists within the same category, the system warns before saving.
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Table 11: Forecast Consumable Demand

Use Case Name:	Forecast Consumable Demand
Scenario:	Consumable stock must be replenished before a shortage disrupts a scheduled session.
Triggering Event:	The laboratory head opens the dashboard and the system recalculates the forecast over the historical usage and consumption records.
Brief Description:	When the head opens the dashboard, the system computes a consumption rate for each consumable from the issued and consumed quantities recorded against past sessions, derives a reorder point from that rate, and compares projected consumption against the sessions remaining in the term as recorded in the laboratory schedule. Items projected to deplete before a scheduled session requires them are listed with a projected depletion date and raised to the laboratory head as a low stock alert. Each recalculation is stored with the time at which it was computed, and forecast confidence is bounded by the depth of historical data available at that time.
Actors:	Laboratory Head
Related Use Cases:	Manage Equipment Inventory, View Real-Time Inventory & Dashboard, Manage Laboratory Logs & Schedule
Stakeholders:	Laboratory Head: needs to restock in advance rather than discovering a shortage during a session. Faculty: need sessions supplied without substitution at short notice.

	Students: need the materials required by the experiment to be present.
Preconditions:	Borrowing and consumption records exist for past sessions. The laboratory schedule for the term has been recorded and the requests for the current term are held in the system. Consumables carry a unit of measure and a reorder level.
Postconditions:	Items at risk of depletion are listed with a projected depletion date. A low stock alert is raised to the laboratory head. The forecast is made available to the dashboard insights.
Flow of Activities:	
Actor	System
1. Head opens the dashboard. 2. Head reviews the items listed as at risk of depletion. 3. Head arranges restocking for the items concerned.	1. System aggregates issued and consumed quantities for each consumable across past sessions. 2. System computes the consumption rate and derives the reorder point. 3. System compares projected consumption against the sessions remaining in the term. 4. System lists the items projected to deplete and raises a low stock alert.
Exception Conditions:	If insufficient historical data exists, the system reports the consumption rate as indicative and suppresses the depletion date. If a consumable has no recorded consumption, it is excluded from the forecast. If the term schedule has not been recorded, the forecast is computed on consumption rate alone without a session horizon.

Table 12: Track Request Status

Use Case Name:	Track Request Status
Scenario:	A requester needs to know the progress of a submitted laboratory service request.
Triggering Event:	The requester enters a request identifier on the request tracking screen.
Brief Description:	The requester enters the request identifier and the system displays the status as Pending, Approved, Declined, Ongoing or Completed together with the details held on the record. The screen is reachable from any device and from outside the university network, and requires no account.
Actors:	Student, Faculty
Related Use Cases:	File Laboratory Service Request, Approve Service Request, Issue Equipment
Stakeholders:	Students and Faculty: need progress visible without visiting the laboratory or emailing the head. Laboratory Head: needs routine status enquiries answered without personal attendance.
Preconditions:	A request exists in the system. The requester holds the request identifier issued on submission.
Postconditions:	The current status and the details held on the record are displayed. No stored record is changed.
Flow of Activities:	
Actor	System
1. Requester opens the request tracking screen. 2. Requester enters the request identifier. 3. Requester reviews the status of the record.	1. System accepts the request identifier. 2. System retrieves the request and its current status. 3. System displays the status and the details held on the record.
Exception Conditions:	If the identifier does not match a request, the system reports that no request was found. If no identifier is entered, the system prompts for

	one. If the request is completed and archived, the status is shown as Completed together with the session date.
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Table 13: Virtual Assistant Query

Use Case Name:	Query Virtual Assistant
Scenario:	A student or faculty member needs an answer about the laboratory outside office hours.
Triggering Event:	A user submits a question to the virtual assistant.
Brief Description:	The assistant answers questions on item availability, laboratory schedule availability, and operating hours, composing its answer from the laboratory's own records rather than from general knowledge. The assistant is reachable from outside the university network. Reservations continue to be filed at the laboratory kiosk and are not processed through the assistant.
Actors:	Student, Faculty
Related Use Cases:	Track Request Status, Manage Equipment Inventory, Manage Laboratory Logs & Schedule
Stakeholders:	Students and Faculty: need routine answers without visiting the laboratory or waiting for office hours. Laboratory Head: needs the volume of repetitive enquiries reduced. NU Fairview: requires access restricted to students and faculty already recorded in the laboratory's records.
Preconditions:	The assistant is reachable and the knowledge base is populated.
Postconditions:	An answer grounded on laboratory records is returned. The exchange is written to the conversation log. No reservation or borrowing record is created.

Flow of Activities:	
Actor	System
1. User opens the assistant from the request tracking screen. 2. User submits a question. 3. User reads the answer returned.	1. System interprets the question and retrieves the relevant laboratory records. 2. System composes an answer grounded on those records. 3. System writes the exchange to the conversation log.
Exception Conditions:	If the question falls outside scope, such as filing a reservation, the assistant directs the user to the laboratory kiosk. If the model service is unavailable or rate limited, the assistant reports that it is temporarily unavailable. If no matching record exists, the assistant states that the information is not held rather than composing an answer.

Table 14: View Real-Time Inventory & Dashboard

Use Case Name:	View Real-Time Inventory & Dashboard
Scenario:	The laboratory head needs the state of the laboratory summarised on signing in.
Triggering Event:	The head signs in and the dashboard metrics load.
Brief Description:	The dashboard summarises inventory totals, low stock items, requests in progress, consumables used within the term, faculty laboratory usage and recent record activity. From these metrics the system generates a written summary so that patterns in usage and resourcing surface without the head interpreting the raw figures.
Actors:	Laboratory Head
Related Use Cases:	Authenticate User, Forecast Consumable Demand, Generate End-Term Report
Stakeholders:	Laboratory Head: needs to act on usage patterns without interpreting raw figures. NU Fairview: benefits from resourcing decisions

	grounded in recorded activity.
Preconditions:	The head is signed in. Records exist for the current term.
Postconditions:	The dashboard metrics are displayed. A written insight summary is generated alongside them. No stored record is changed.
Flow of Activities:	
Actor	System
1. Head signs in to the system. 2. Head reviews the dashboard metrics. 3. Head reads the generated insight summary.	1. System aggregates inventory, request, consumption and usage figures for the term. 2. System renders the metrics on the dashboard. 3. System generates a written summary of the patterns in those metrics. 4. System displays low stock items and forecast warnings alongside the summary.
Exception Conditions:	If insufficient data exists for the term, the metrics are shown and the generated summary is suppressed. If the model service is unavailable, the metrics are displayed without the generated summary.

Table 15: Generate End-Term Report

Use Case Name:	Generate End-Term Report
Scenario:	The term has ended and the laboratory must report its activity to the administration.
Triggering Event:	The laboratory head triggers the end-of-term report.
Brief Description:	The system compiles laboratory facility usage per faculty member, drawn from the usage entry recorded for each completed session whether or not equipment was issued, together with the most borrowed equipment, the totals of broken, lost and consumed items, and the waste disposal entries with their hazardous classification, and exports the

	result to the laboratory head for submission. The figures are generated directly from stored records rather than from a hand-built formula set.
Actors:	Laboratory Head
Related Use Cases:	View Real-Time Inventory & Dashboard, Log Waste Disposal, Process Clearance, Manage Laboratory Logs & Schedule
Stakeholders:	Laboratory Head: needs reporting that does not depend on preserving a spreadsheet formula set. NU Fairview administration: receives the figures required at the end of each term. Faculty: are reported against their recorded laboratory usage.
Preconditions:	The term is recorded in the system and its records are complete, and the usage entries for the term have been generated. The head is signed in.
Postconditions:	The report is compiled and exported to the laboratory head. Every figure is traceable to the stored records behind it. The head submits the exported report to the administration; the system stores the records the report is drawn from rather than the report itself.
Flow of Activities:	
Actor	System
1. Head opens the reporting screens for the term. 2. Head selects the term and triggers the report. 3. Head exports the report for submission.	1. System aggregates usage per faculty member and per item across the term. 2. System compiles the broken, loss and consumed totals and the waste disposal entries. 3. System renders the report with hazardous items shown separately. 4. System exports the report in the selected format.
Exception Conditions:	If clearances remain open at the end of the term, they are listed in the report as outstanding. If the term holds no records, an empty report is produced with a notice to that effect.

	If the export fails, the head is notified and may retry.
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Table 16: Manage Laboratory Logs & Schedule

Use Case Name:	Manage Laboratory Logs & Schedule
Scenario:	The Dean's term schedule must be recorded so that rooms can be booked and vacant blocks released.
Triggering Event:	The laboratory head records the term schedule.
Brief Description:	The head records, for each laboratory room, which time blocks are committed to laboratory sessions, which are committed to lecture sessions and which remain vacant, against the term to which the schedule belongs. Blocks the schedule leaves vacant become available for out-of-schedule requests. The Physics and Circuits laboratory logs are maintained on their own screens, where the laboratory head records the usage of each completed session once its return has been reconciled, listing the items used and consumed where equipment was issued and recording the session alone where it was facility use only, and the system compiles from those entries the average equipment use, the most used items and the usage per professor. The head also records here any broken, loss or consumed adjustment that did not arise from a borrowing transaction. These compiled figures supply the usage reported at the end of the term.
Actors:	Laboratory Head
Related Use Cases:	File Laboratory Service Request, Forecast Consumable Demand, Generate End-Term Report, Manage Daily Task Log
Stakeholders:	Laboratory Head: needs one schedule per room in place of separate weekly grids and colour coding. Dean: prepares the term schedule and requires it applied accurately. Students and Faculty: rely on published vacant

	blocks when filing out-of-schedule requests.
Preconditions:	The term schedule has been received from the Dean's office. The head is signed in.
Postconditions:	The laboratory schedule is updated for each room and is recorded against the term. Vacant blocks are published and become available for out-of-schedule requests. A usage entry is recorded in the Physics or Circuits laboratory log for each completed session and made available to reporting. Any manual broken, loss or consumed adjustment is written to the Broken, Loss and Consumed log without a borrowing reference.
Flow of Activities:	
Actor	System
1. Head opens Laboratory Logs and Schedule. 2. Head records the committed blocks for each room from the Dean's term schedule. 3. Head marks the remaining blocks vacant. 4. Head records any broken, loss or consumed adjustment that did not arise from a borrowing transaction, such as an item damaged during maintenance or found missing at an inventory count.	1. System displays the weekly grid for the selected room. 2. System validates that committed blocks do not overlap. 3. System stores the schedule and publishes the vacant blocks. 4. System compiles the recorded usage entries and makes them available to reporting.
Exception Conditions:	If two committed blocks overlap, the system rejects the entry. If a block already carries an approved reservation, it cannot be recommitted until that reservation is released. If the term schedule has not yet been received, rooms remain unavailable for on-schedule requests.

Table 17: Manage Daily Task Log

Use Case Name:	Manage Daily Task Log
Scenario:	The laboratory head records the day's work at the close of the working day.

Triggering Event:	The laboratory head completes and submits the daily task report.
Brief Description:	The head records work availability for the day, indicating whether the post was worked on site, worked from home, taken as a day off or fell on a holiday, together with the morning and afternoon reviews completed, the email requests and inquiries handled, the monthly summary, the dispensing and receiving of equipment, any overtime and any remarks. The entry is stored against the date and supplies part of the figures reported at the end of the term.
Actors:	Laboratory Head
Related Use Cases:	Manage Laboratory Logs & Schedule, Generate End-Term Report
Stakeholders:	Laboratory Head: needs the daily report completed without maintaining a separate form. NU Fairview: requires a record of laboratory staffing and daily activity.
Preconditions:	The head is signed in. The sessions for the day have concluded.
Postconditions:	A daily task entry is stored against the date and marked on-time or late according to when it was submitted. The entry is made available to the end-term report.
Flow of Activities:	
Actor	System
1. Head opens the Daily Task screen. 2. Head records work availability for the day. 3. Head records the reviews completed, the inquiries handled and the remaining entries for the day. 4. Head submits the daily task report.	1. System displays the daily task form for the current date. 2. System validates that no entry already exists for that date. 3. System stores the entry. 4. System makes the entry available to reporting.
Exception Conditions:	If an entry already exists for the date, the head amends it rather than creating a duplicate. If the day is recorded as a day off or a holiday, the remaining entries are not required.

	If the report is submitted after the date has passed, it is stored against the actual date and flagged as late.
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3.1.5 Gap Analysis

Table 18 sets out the gap analysis, mapping each product backlog item to the current system and to the proposed change.

Table 18: Gap Analysis

Product Backlog ID	Current System	Proposed Changes	Impact
01	Inventory is held in the inventory finder worksheet as name, description, existing quantity, unit and type, with no condition field and no record of how much of each item is currently out or committed; edits are typed directly into cells.	Provide an inventory module supporting add, edit and remove, in which each holding is recorded by name with a condition status and quantities tracked as existing, available, in use and reserved.	Stock levels become visible at a glance and changes are attributable rather than silently overwritten.
02	Borrowing is recorded in the Physics and Circuits usage logs by date, subject code, professor, particulars and quantity; no student is named anywhere in the workbook.	Provide a borrowing record capturing student ID numbers and time against the approved request.	Accountability attaches to the borrower rather than only to the supervising professor.
03	Damage and loss are logged by date, professor, item, quantity and unit; the accountable group is traceable only through the signed paper slip for that session.	Record damage and loss directly against the accountable group drawn from the borrowing record.	Removes the paper cross-reference step and prevents unattributed losses.
04	Availability is established by opening the inventory finder and	Provide a live availability view showing quantities on	Availability can be checked without a physical count.

	reading a quantity that is updated by hand after each session.	hand and expected return times.	
05	End-of-term figures are computed by formula blocks in the Broken, Loss and Consumed worksheet and exported manually into a presentation.	Generate summaries and transaction logs directly from stored records.	Reporting no longer depends on preserving a hand-built formula set.
06	Clearance is issued on paper, signed by the student, a faculty member and the laboratory head, then forwarded to the registrar.	Provide a digital clearance record raised and tracked by the laboratory head.	Removes lost and incomplete clearance records and shortens processing time.
07	Required equipment is written onto the paper service request form and re-encoded by the laboratory head.	Allow equipment to be selected from the catalogue at the kiosk with live stock indicators.	Removes re-encoding and prevents requests for items that are unavailable.
08	Availability is checked against weekly grids kept per laboratory room; a grid for the Circuits room does not yet exist.	Provide a calendar with per-room conflict detection covering every laboratory room.	Conflicts are prevented at filing rather than discovered afterwards.
09	Approval is a signature on a printed form; the signed copy is stored on the university SharePoint service and reached from the request register through a stored link.	Record approval through a confirmation control in an automated email, requiring no account.	Removes physical routing of forms and the dependency on external file storage.
10	Out-of-schedule use is arranged in person or by email against vacant blocks. The two-day advance requirement is applied by the laboratory head but is not recorded or	Allow out-of-schedule requests to be filed against vacant blocks with the two-day requirement enforced by the system.	Removes on-the-spot requests and makes the advance requirement auditable.

	enforced anywhere in the workbook.		
11	Recurrence is written as free text in the remarks column of the request register, for example EVERY SATURDAY.	Type each request as whole-term or one-time and handle the request lifecycle accordingly.	Completion counts are no longer distorted by recurring requests.
12	The usage logs carry a quantity column and a consumed column with a type marker, but nothing prevents an equipment item from being marked consumed.	Record issued and returned quantities separately and bar equipment from being marked consumed.	Consumed counts reflect genuine depletion only, and unreturned equipment is recorded as broken or lost.
13	The hazardous waste disposal worksheet exists as a prepared template in which no disposal has been recorded.	Provide a disposal log linked to the inventory, recording reason, quantity, unit, date and classification.	Withdrawals from the laboratory become traceable and hazardous items become reportable.
14	Consumable levels are known only from the last manual decrement, so a shortage surfaces when a scheduled session already requires the item.	Forecast consumable demand from historical usage and raise stockout alerts ahead of scheduled sessions.	Restocking becomes anticipatory rather than reactive.
15	Item availability, operating hours and request status are answered by the laboratory head in person or by email during office hours.	Provide a virtual assistant answering from laboratory records to verified users at any hour.	Reduces repetitive workload and removes the office-hours constraint on routine inquiries.
16	Usage figures sit in formula blocks and are interpreted by the laboratory head.	Generate automated insights from the dashboard metrics.	Patterns in usage and resourcing surface without manual interpretation.

3.1.6 Context Diagram

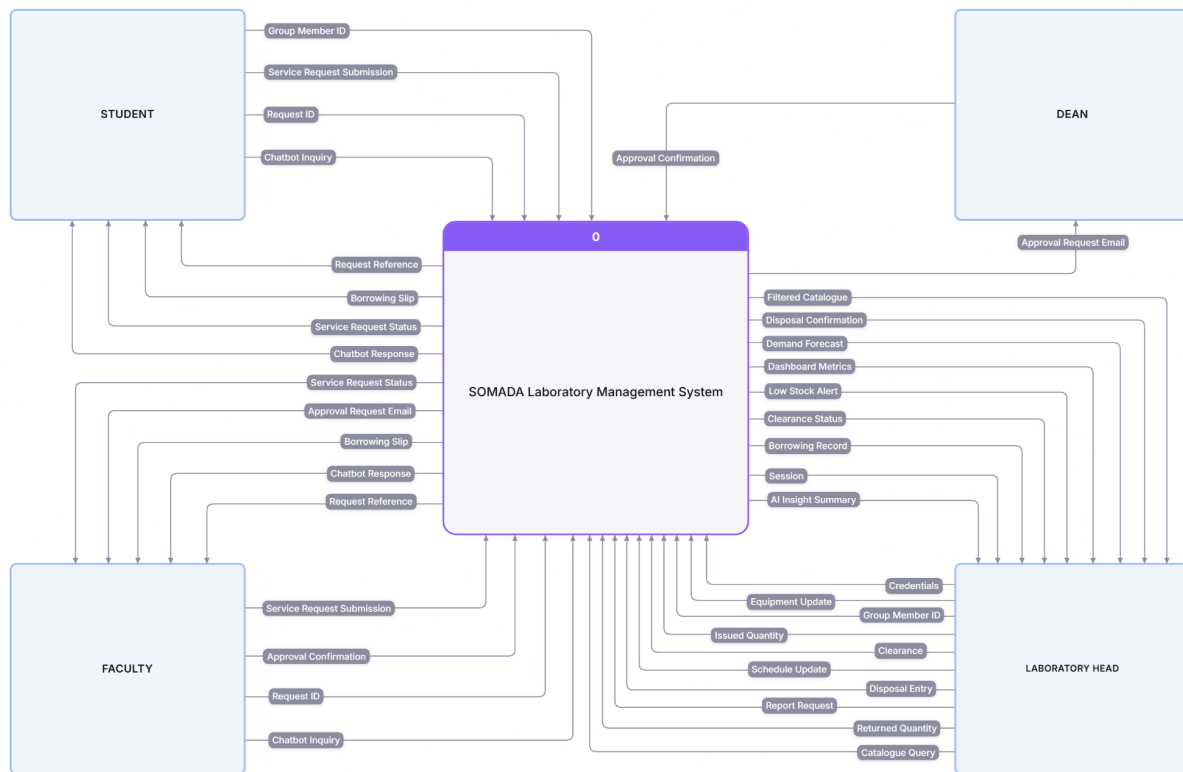


Figure 2: Context Diagram

Figure 2 presents the context diagram, showing SOMADA as a single process together with the four external entities it exchanges data with. Students and faculty submit service requests and inquiries and receive references, statuses, borrowing slips and assistant responses; faculty and the Dean receive approval request emails and return confirmations; and the laboratory head supplies inventory, issuance, return, clearance, schedule and disposal entries and receives the catalogue, alerts, dashboard metrics, the demand forecast and the exported end-term report.

3.1.7 Data Flow Diagrams

Level 1

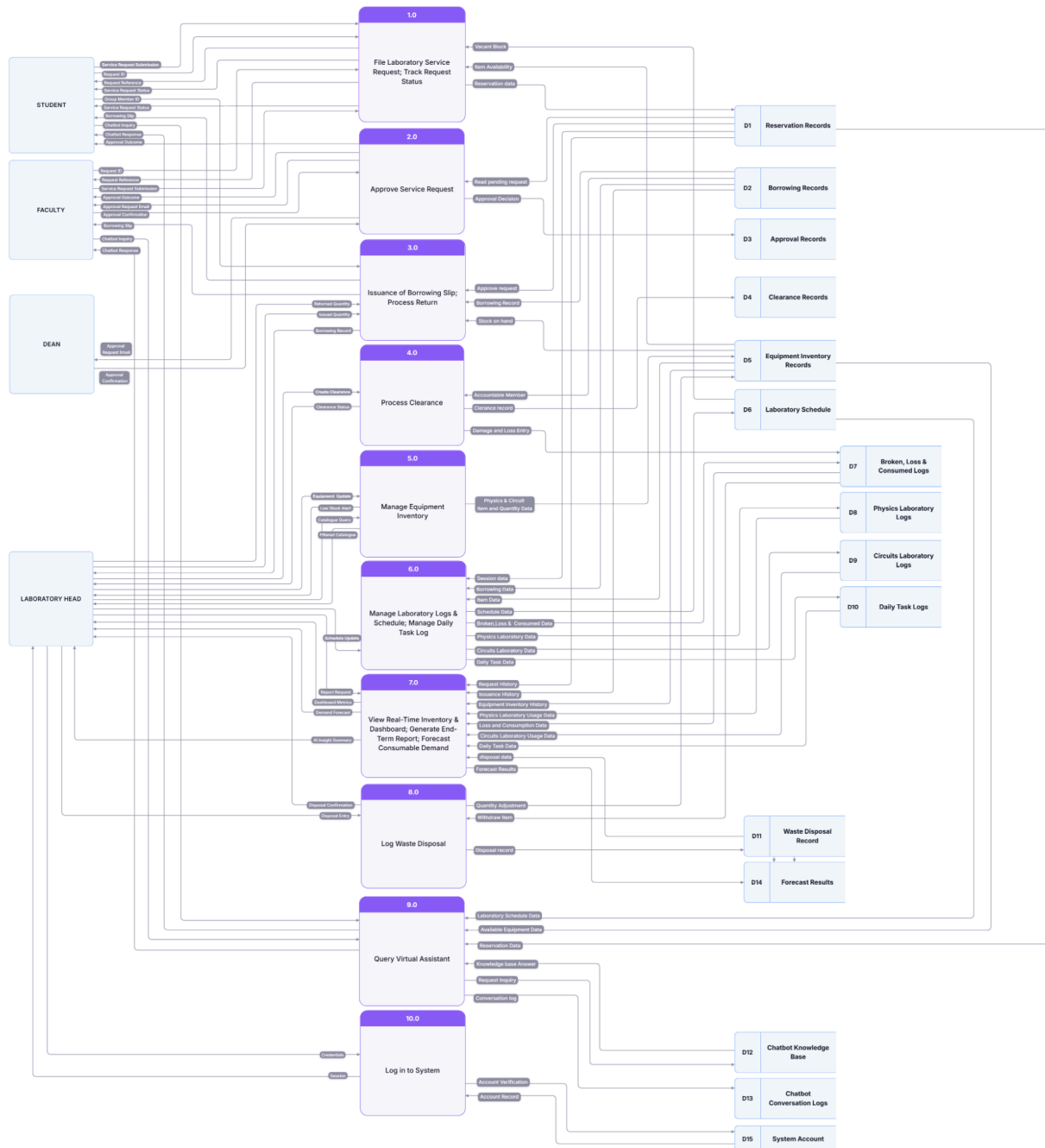


Figure 3: Level 1 Data Flow Diagram

Figure 4: Level 2 Data Flow Diagram — Process 1.0 File Laboratory Service Request; Track Request Status

Process 2.0 - Approve Service Request

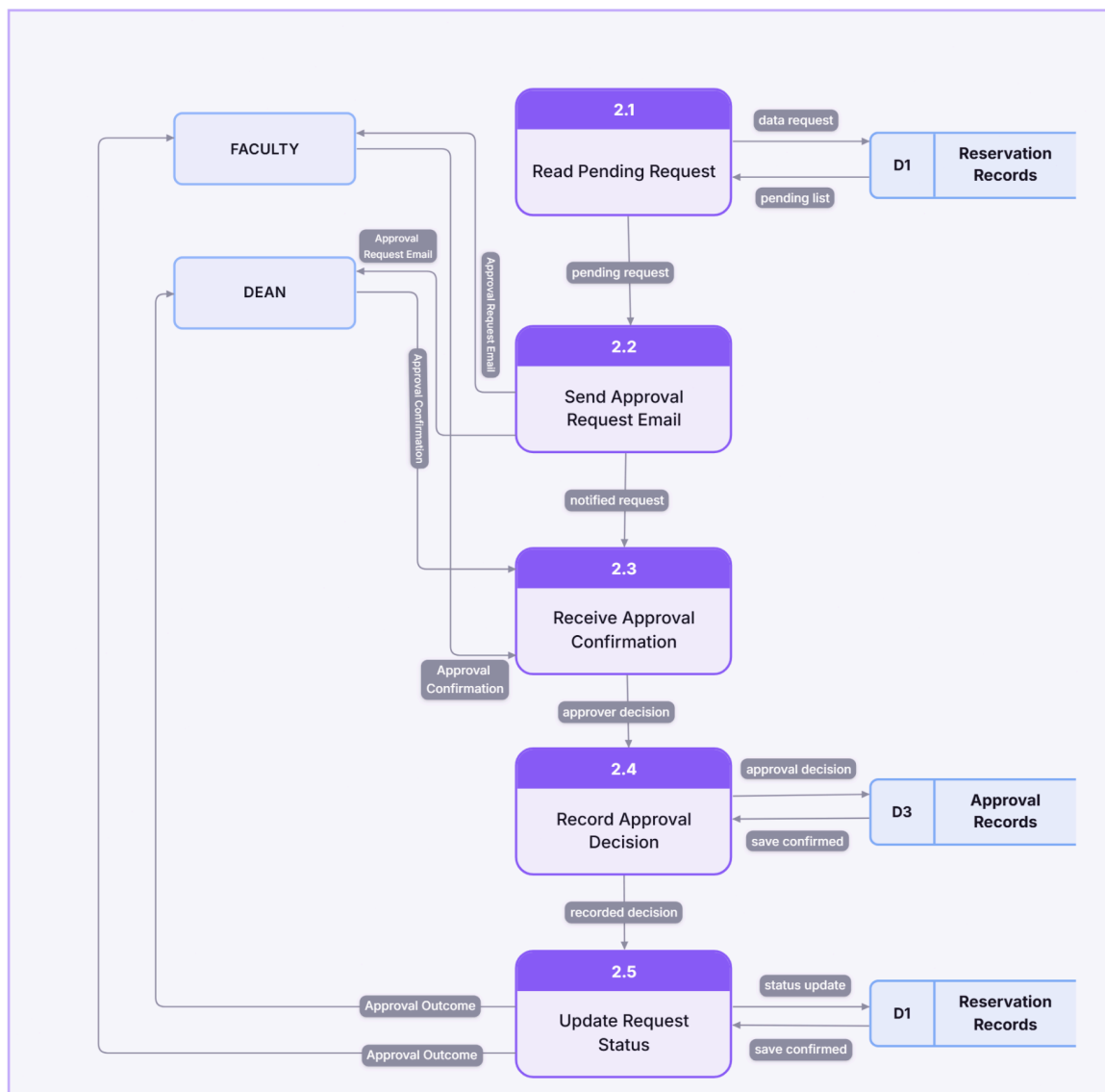


Figure 5: Level 2 Data Flow Diagram — Process 2.0 Approve Service Request

Process 3.0- Issuance of Borrowing Slip; Process Return

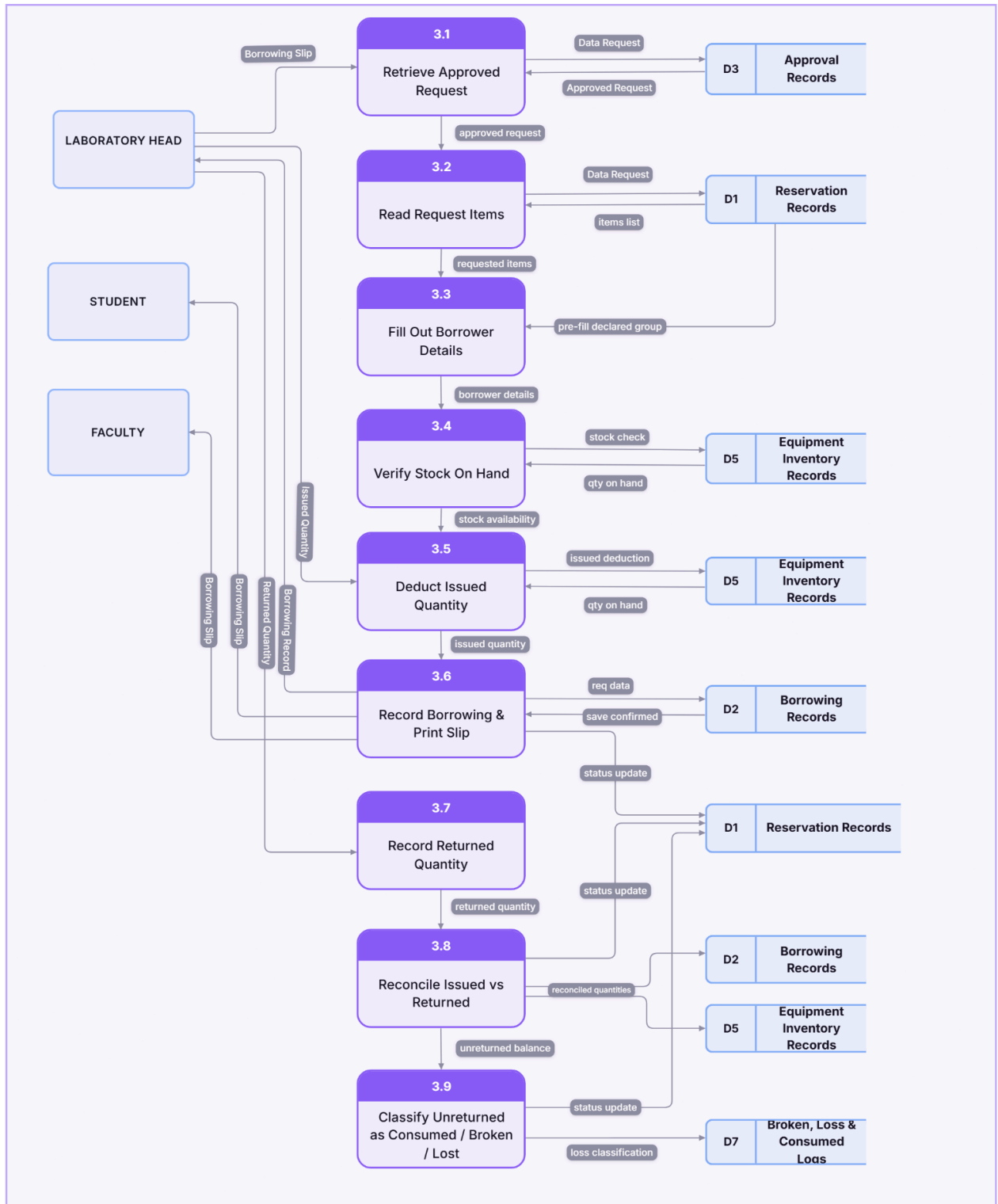


Figure 6: Level 2 Data Flow Diagram — Process 3.0 Issue Equipment; Process Return

Process 4.0 - Process Clearance

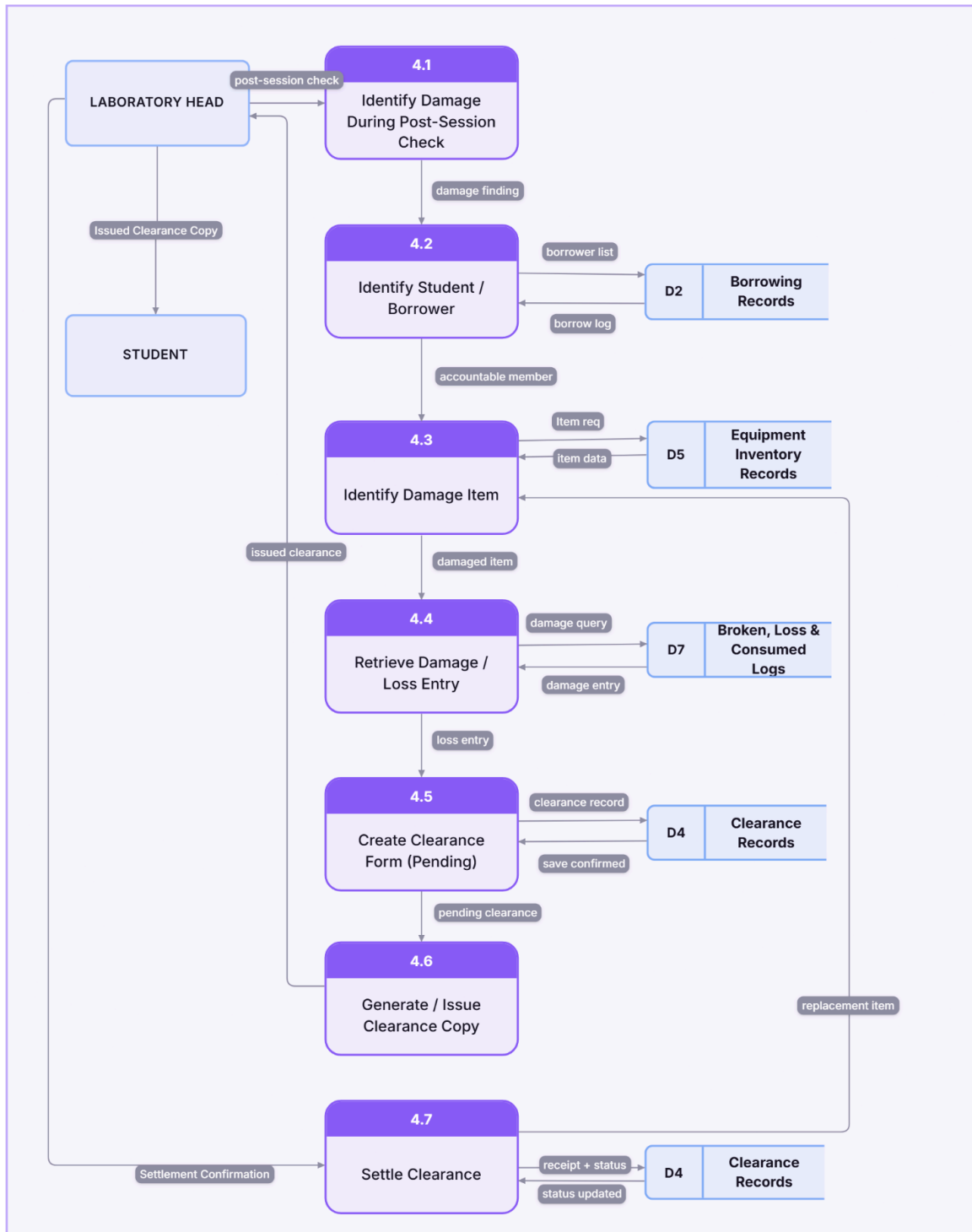


Figure 7: Level 2 Data Flow Diagram — Process 4.0 Process Clearance

Process 5.0 - Manage Equipment Inventory

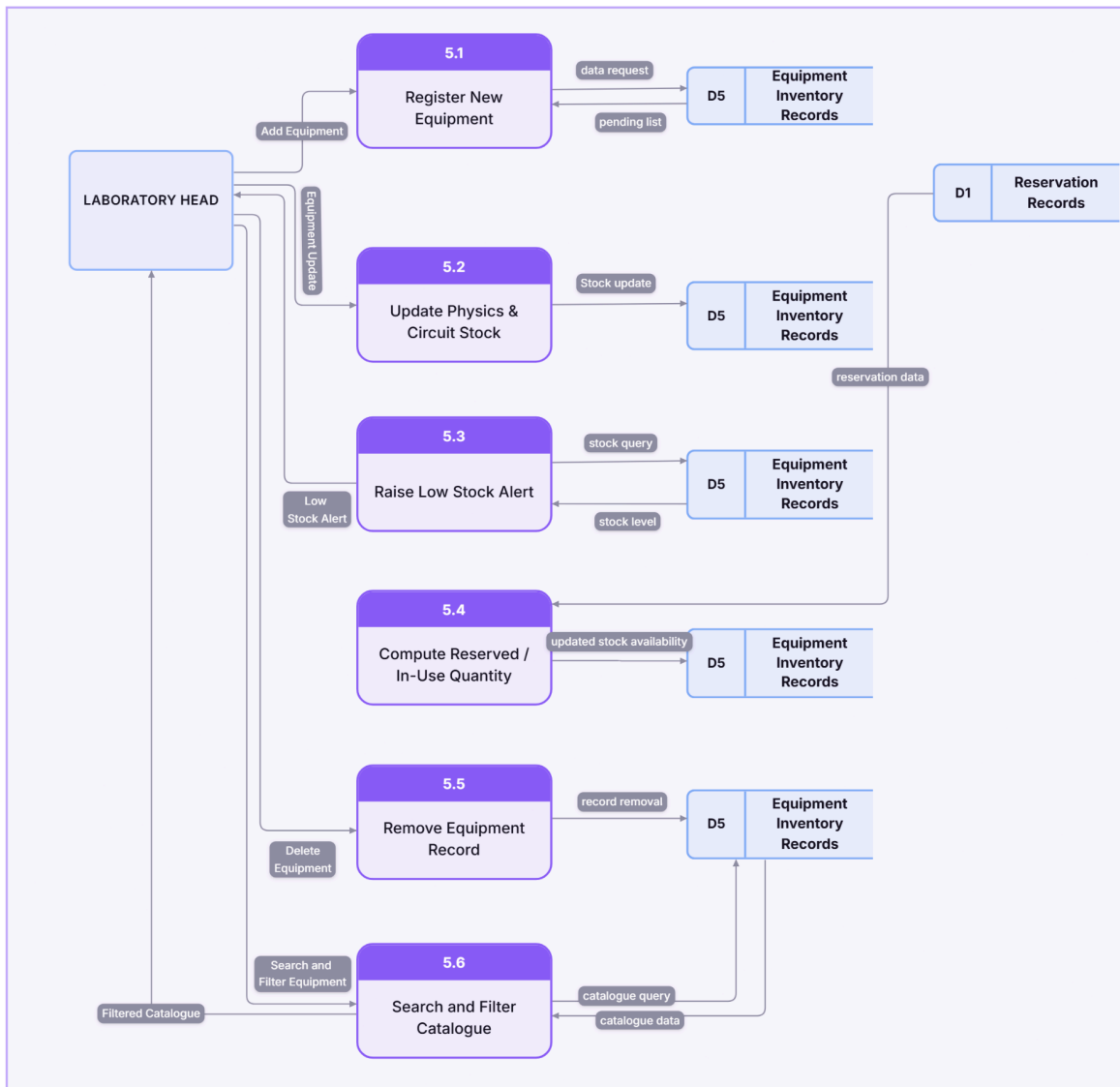


Figure 8: Level 2 Data Flow Diagram — Process 5.0 Manage Equipment Inventory

Process 6.0 - Manage Laboratory Logs & Schedule; Manage Daily Task Log

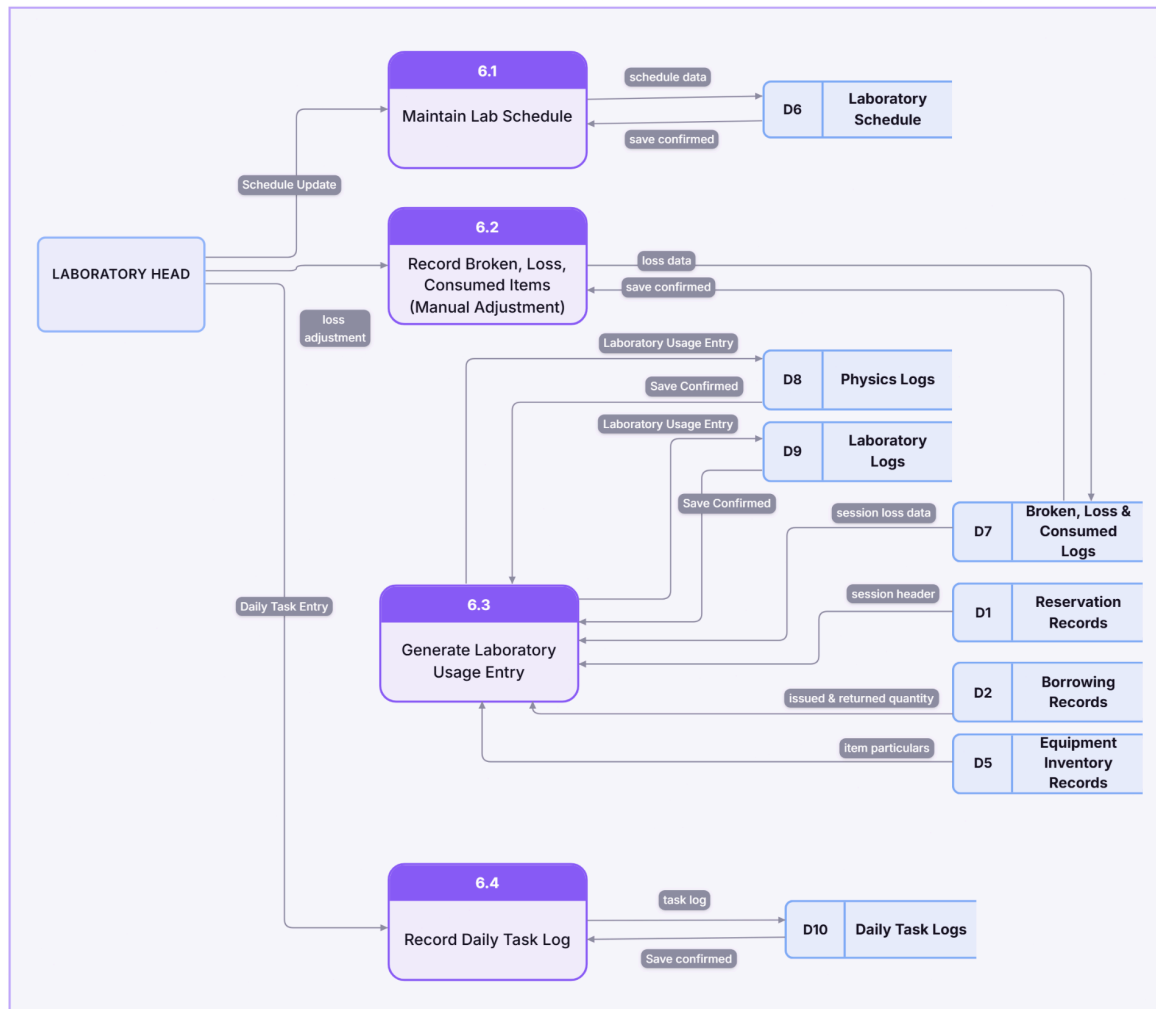


Figure 9: Level 2 Data Flow Diagram — Process 6.0 Manage Laboratory Logs & Schedule; Manage Daily Task Log

**Process 7.0 - View Real-Time Inventory & Dashboard; Generate End-Term Report;
Forecast Consumable Demand**

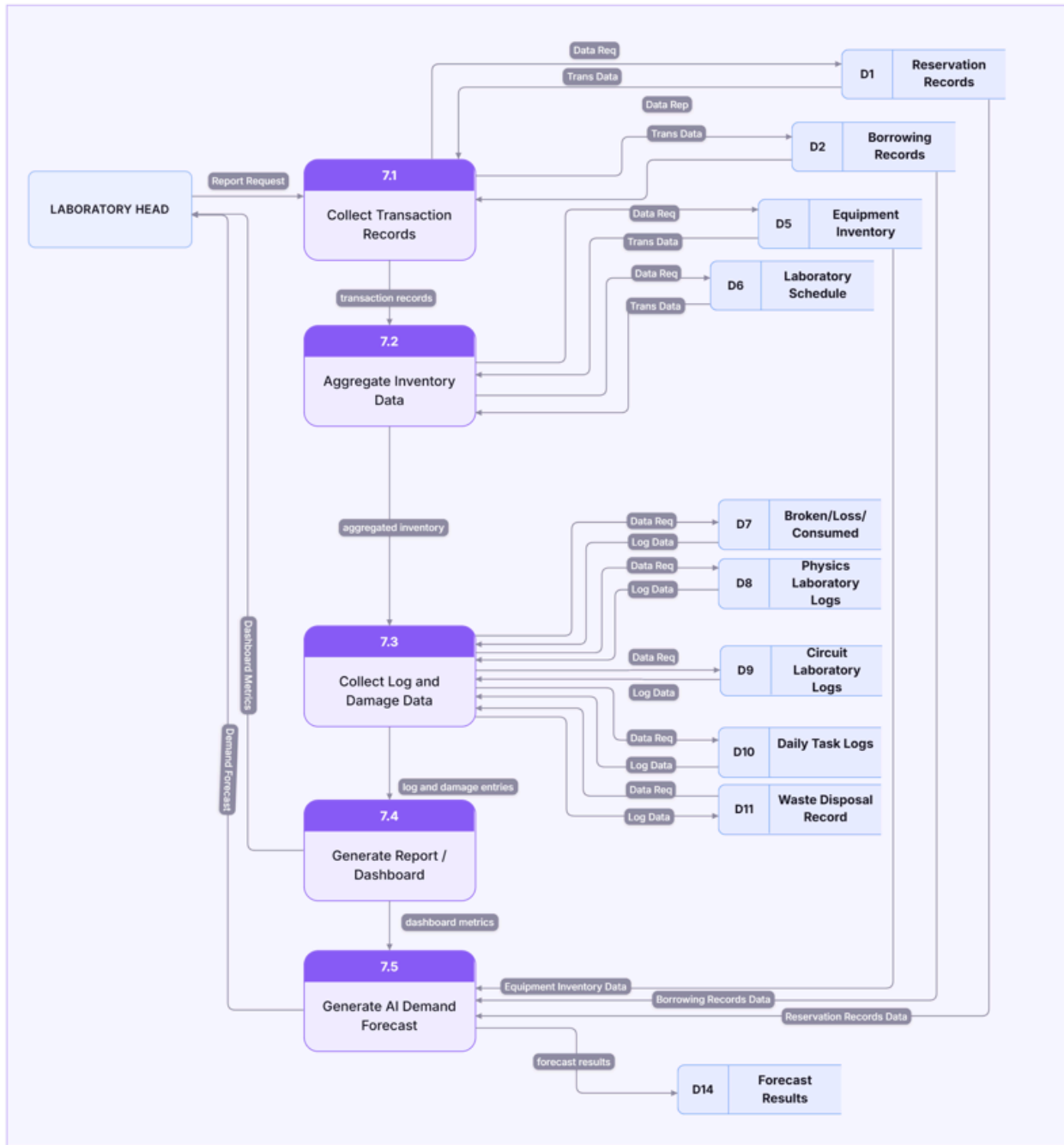


Figure 10: Level 2 Data Flow Diagram — Process 7.0 View Real-Time Inventory & Dashboard; Generate End-Term Report; Forecast Consumable Demand

Process 8.0 - Log Waste For Disposal

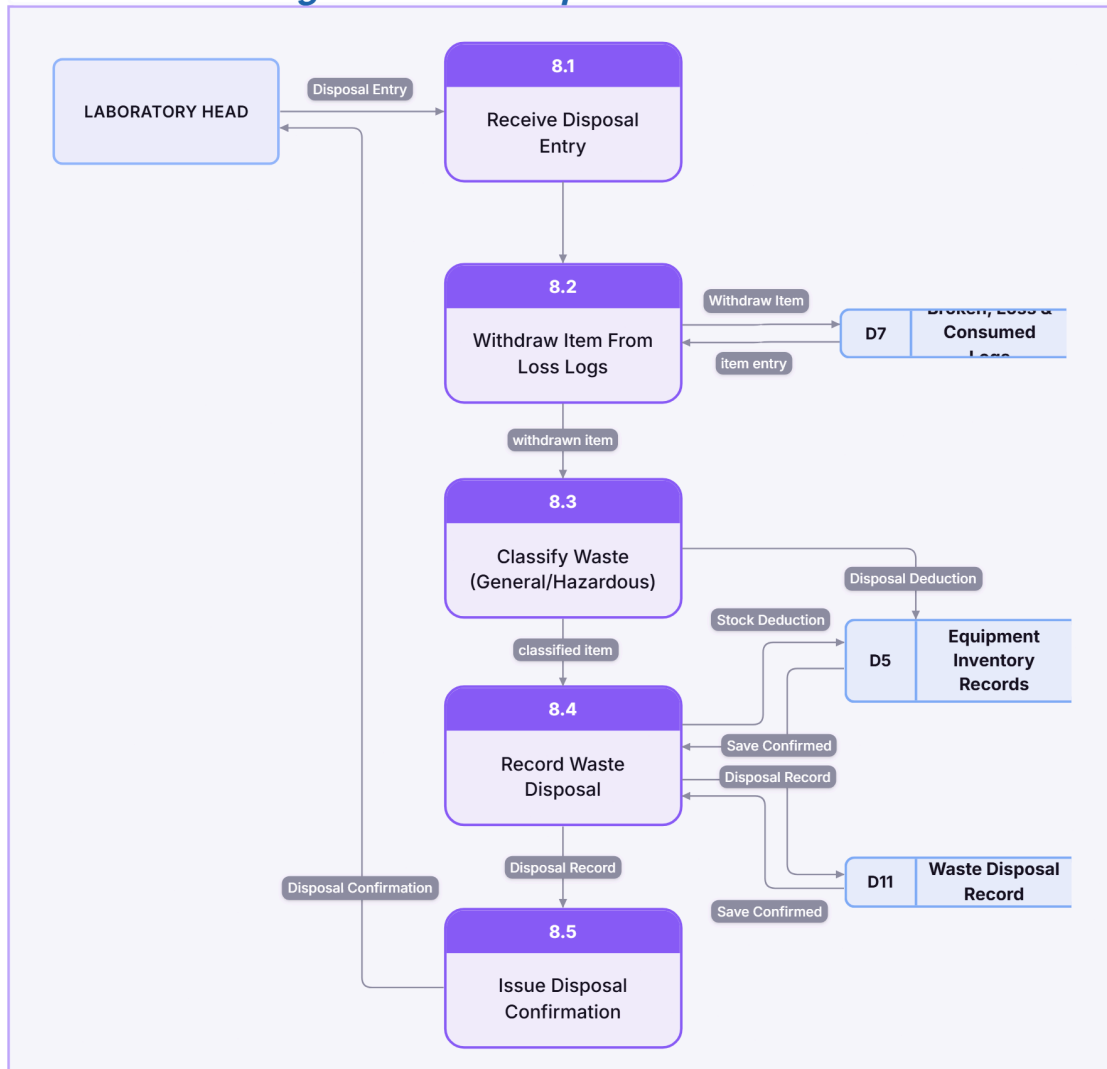


Figure 11: Level 2 Data Flow Diagram — Process 8.0 Log Waste Disposal

Process 9.0- Virtual Assistant Query

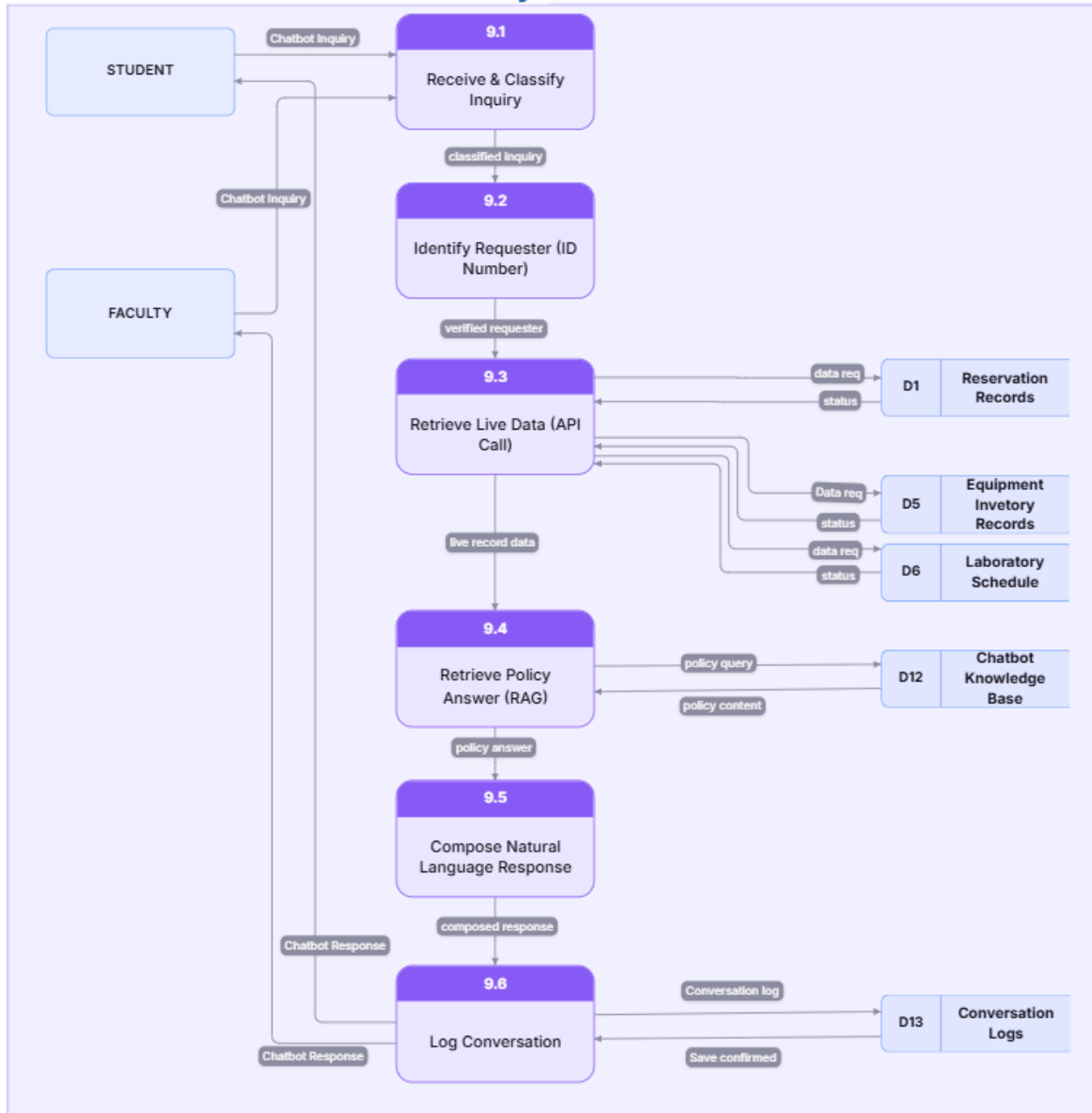


Figure 12: Level 2 Data Flow Diagram — Process 9.0 Query Virtual Assistant

Process 10.0 - Login to System

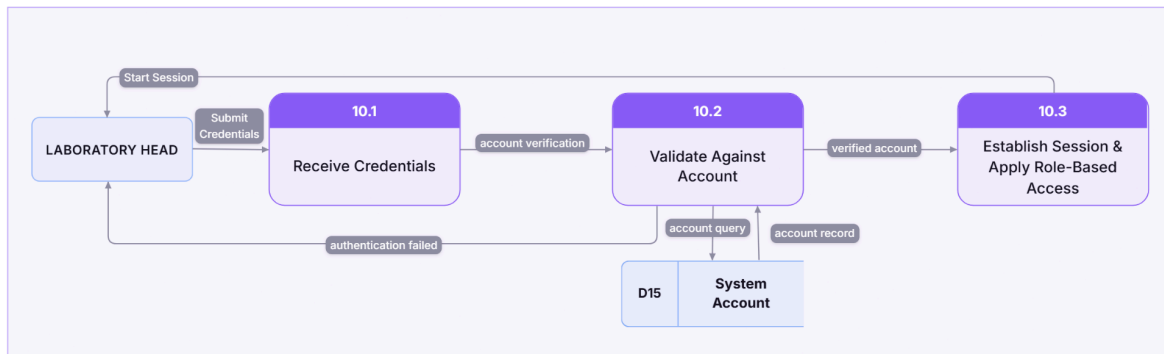


Figure 13: Level 2 Data Flow Diagram — Process 10.0 Authenticate User

3.1.8 Entity-Relationship Diagram

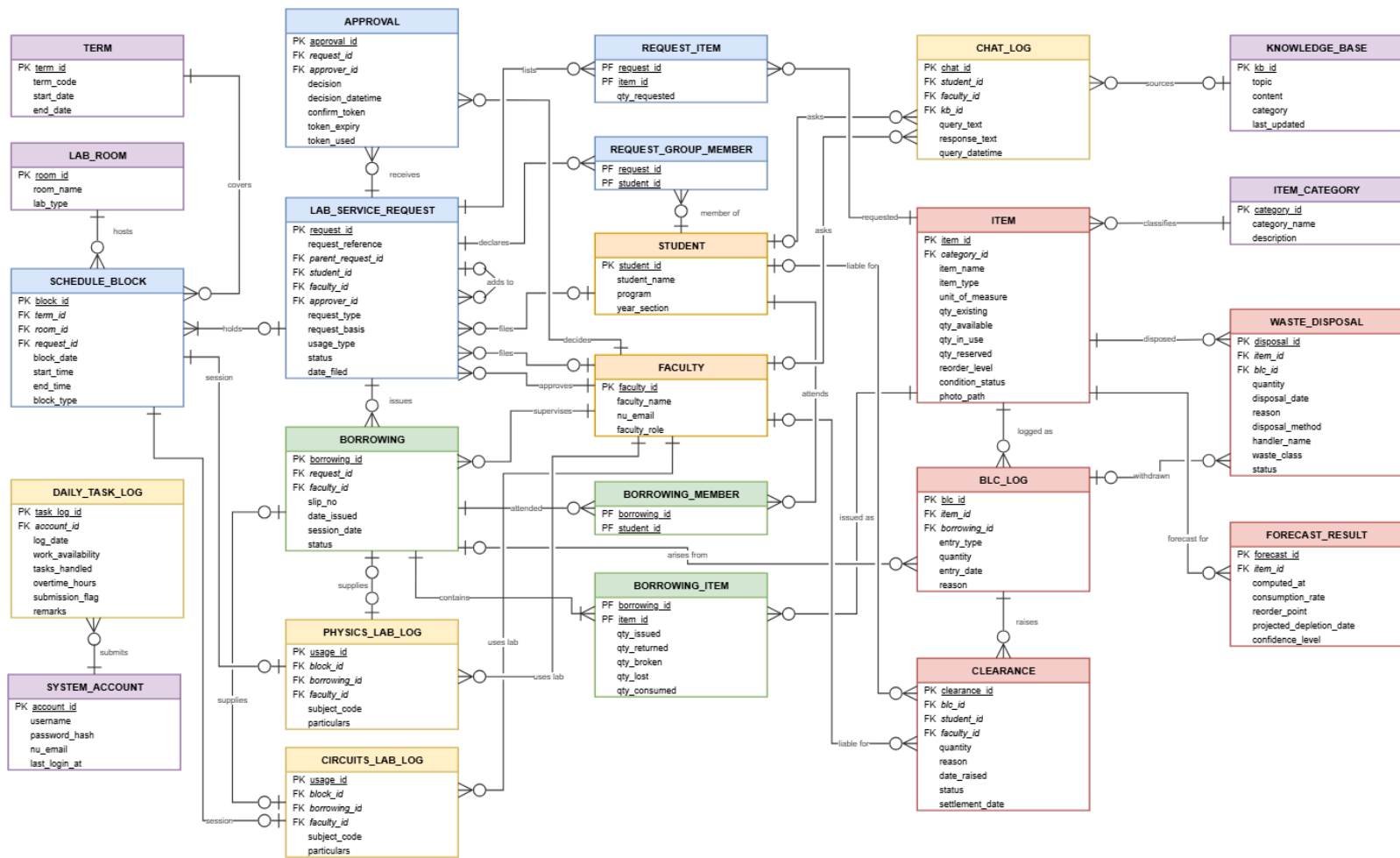


Figure 14: Entity Relationship Diagram